

CYC Standard Data Format

What is a CYC file?

A CYC (Cycle Counting) file contains the counted fatigue cycles obtained from an LDS file using a selected cycle counting method.

File naming convention

File: CYC_{ResearcherLastname}_{YYYY-MM}_{method}.csv

- YYYY-MM = starting month of the experiment
- method = range_mean, range_pair, simplified_rainflow, etc.
- Example: CYC_Dupont_2024-03_range_mean.csv

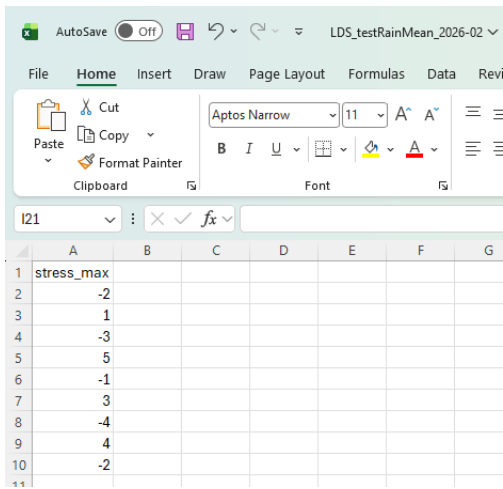
File format requirements

- Encoding: UTF-8
- Separator: comma (,)
- File type: .csv
- Header required
- Units: MPa

Required columns (column names must be exact)

Variable name	Description	Symbol	Unit	Data type	Mandatory
stress_range	Stress range (= 2 * sigma_amplitude)	delta_sigma	[MPa]	double	y
stress_mean	Mean stress	sigma_mean	[MPa]	double	y
stress_ratio	Stress ratio	R	[-]	double	y
n_cycles	Number of cycles	n		double	y
cum_n_cycles	cumulative number of cycles for a given bin of sigma_a, sigma_mean and R.			double	y

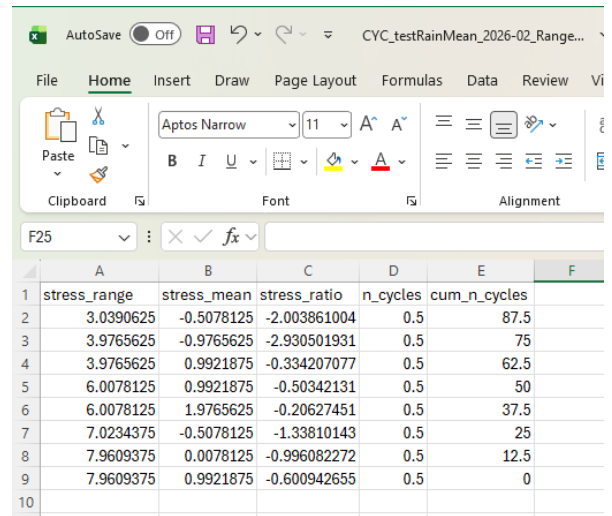
Example of valid CYC file



	A	B	C	D	E	F	G
1	stress_max						
2		-2					
3		1					
4		-3					
5		5					
6		-1					
7		3					
8		-4					
9		4					
10		-2					

LDS file

Range Mean



	A	B	C	D	E	F
1	stress_range	stress_mean	stress_ratio	n_cycles	cum_n_cycles	
2	3.0390625	-0.5078125	-2.003861004	0.5	87.5	
3	3.9765625	-0.9765625	-2.930501931	0.5	75	
4	3.9765625	0.9921875	-0.334207077	0.5	62.5	
5	6.0078125	0.9921875	-0.50342131	0.5	50	
6	6.0078125	1.9765625	-0.20627451	0.5	37.5	
7	7.0234375	-0.5078125	-1.33810143	0.5	25	
8	7.9609375	0.0078125	-0.996082272	0.5	12.5	
9	7.9609375	0.9921875	-0.600942655	0.5	0	
10						

CYC file

For the CSV file, Microsoft Excel works well for editing and saving the data.

However, other tools such as LibreOffice Calc, Google Sheets, or any text editor (e.g., Notepad++, VS Code) can also be used, provided the file is saved in CSV format (UTF-8 encoding).